

Reviewer Pack — Minimal Rank of Quantum Group Representations and Resolution...

Entry f746dc96472a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quantum Group Representations and Resolution Proof Width Correlation

Entry ID: f746dc96472a **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-02 10:07:27 UTC

1. Conjecture

Field A (mathematical branch): Quantum Group Theory **Field B** (complexity object): Resolution Proof Complexity

Statement:

For every resolution proof with width w , the minimal rank of a quantum group representation associated with its clause set is upper bounded by w^2 .

Rationale (proposer's reasoning):

Quantum groups provide a framework that captures certain algebraic structures underlying complexity. The minimal rank of a quantum group representation could serve as an invariant for the hardness of a resolution proof, potentially revealing hidden structures in proof complexity.

Taxonomy category: QuantumGroupTheory × ResolutionProofComplexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5dad23de8414c3c9

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a given resolution proof width w , the minimal rank of the quantum group representation associated with its clause set will be considered supported if the average minimal rank across all seeds is less than or equal to w^2 AND at least 80% of the seeds meet this condition.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	0.95	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "quantum group theory" AND "resolution proof complexity" AND minimal rank" - "minimal rank of quantum group representations" AND resolution proof width correlation" - "quantum group representation" AND upper bound w^2 resolution proof width"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(width, n):
        cnf = []
        for _ in range(n):
            clause = [random.randint(1, width * 2) for _ in range(random.randint(1, width))]
            cnf.append(clause)
        return cnf

    def compute_minimal_rank(cnf):
        # Simplified computation of minimal rank (placeholder)
        return len(cnf) ** 0.5

    n_max = 40
    instances_tested = 30
    total_rank = 0

    for n in range(5, n_max + 1, 5):
        cnf = generate_cnf(n, instances_tested)
        rank = compute_minimal_rank(cnf)
        total_rank += rank

    metric_value = total_rank / (n_max // 5)

    conjecture_holds = metric_value <= n_max ** 2
    counterexample = "" if conjecture_holds else f"mean_rank={metric_value}, max_n={n_max}"

    return {
        "metric_name": "minimal_rank",
```


30 instances tested, which is insufficient to draw conclusions about the conjecture’s validity.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: The test results indicate that for all trials, the conjecture holds true with an average minimal rank of 5.477225575051661, which is less than or equal to 6 | next: Further investigation is needed to validate the critic’s concerns regarding the accuracy of the placeholder computation and the sufficiency of the test range and number of instances. Consider increasing the range of n_max and the number of instances tested to strengthen the evidence for the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	23718
2	preregistration	ollama_remote	glm4:latest	0	0	9357
3	novelty	ollama_remote	glm4:latest	0	0	9060
4	novelty	ollama_remote	glm4:latest	0	0	16455
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8980
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22715
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11197
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15808
9	critic	ollama_remote	glm4:latest	0	0	14535
10	judge	ollama_remote	glm4:latest	0	0	13207

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 145032 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/f746dc96472a.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/f746dc96472a.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/f746dc96472a.tar.gz (if generated)